

Random Variables

Definition 1.5.1

Consider Ω .

A function X that assigns each $\omega \in \Omega$ exactly one number $X(\omega) = x$ is called a rv.

The space of X is $\mathbb{R}$.

$$\hookrightarrow \mathcal{D} = \{x \mid \exists \omega : X(\omega) = x, \omega \in \Omega\}$$

($\mathcal{D}$ countable set or interval)

Prob. of $X \Rightarrow$ DISTRIBUTION OF X

RV $\begin{cases} \nearrow \text{discrete} \\ \searrow \text{continuous} \end{cases}$

DISCRETE

$$\mathcal{D} = \{d_1, \dots, d_m\}$$

$P_X(d_i)$ on $\mathcal{D}$:

$$P_X(d_i) = P(\{\omega \mid X(\omega) = d_i\}), \quad i = 1, \dots, m$$

Probability mass function

pmf

$$P_X(D) = \sum_{d_i \in D} P_X(d_i), D \subset \mathcal{D}$$

Example 1.5.1

Rolling two dice.

$$\Omega = \{(i, j) \mid 1 \leq i, j \leq 6\},$$

$$P(\{(i, j)\}) = \frac{1}{36}$$

$$X = X(i, j) = i + j$$

$$\text{Space of } X: \mathcal{D} = \{2, \dots, 12\}$$

This gives:

Range value x	2	3	4	5	6	7	8	9	10	11	12
Probability $p_X(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Let:

$$\mathcal{B}_1 = \{x \mid x = 7, 11\}$$

$$\mathcal{B}_2 = \{x \mid x = 2, 3, 12\}$$

$$P_X(B_1) = \sum_{x \in B_1} P_X(x) = \frac{6}{36} + \frac{2}{36} = \frac{8}{36}$$

$$P_X(B_2) = \sum_{x \in B_2} P_X(x) = \frac{1}{36} + \frac{2}{36} + \frac{1}{36} = \frac{4}{36}$$

Continuous

→ $f_X(x)$ s.t. for any interval $(a, b) \in \mathcal{D}$

$$P_X((a, b)) = P(\{\omega \in \Omega \mid a < X(\omega) < b\})$$

$$= \int_a^b f_X(x) dx$$

PROBABILITY THAT X FALLS BETWEEN
 a AND b IS AREA UNDER $f_X(x)$.

$$\rightarrow f_X(x) \geq 0$$

$$\rightarrow P_X(\mathcal{D}) = \int_{\mathcal{D}} f_X(x) dx = 1$$



PROBABILITY DENSITY FUNCTION

(PDF)

Example 1.5.2

Choose $X \in (0,1)$, $r \in \mathbb{R}$

Let X be number chosen.

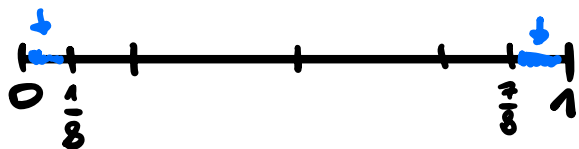
Space of $X = \mathcal{D} = (0,1)$

$$P_X((a,b)) = b-a, \quad 0 < a < b < 1$$

PDF

$$f_X(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Prob of X being less than $\frac{1}{8}$ or greater than $\frac{7}{8}$:



$$P(\{X < \frac{1}{8}\} \cup \{X > \frac{7}{8}\})$$

$$= \int_0^{1/8} dx + \int_{7/8}^1 dx$$

$$= (\frac{1}{8} - 0) + (1 - \frac{7}{8})$$

$$= \frac{1}{4} //$$

NOTE THAT:

DISCRETE NOT POSSIBLE

For any point a : $0 < a < 1$

choose n_0 s.t.:

$$0 < a - \frac{1}{n_0} < a < a + \frac{1}{n_0} < 1$$

$$\Rightarrow \{a\} \subset (a - \frac{1}{n_0}, a + \frac{1}{n_0})$$

$$\Rightarrow P(X=a) \leq P(a - \frac{1}{n} < X < a + \frac{1}{n})$$

$$= \cancel{a} + \frac{1}{n} - (\cancel{a} + \frac{1}{n})$$

$$= \frac{2}{n}, \forall n \geq n_0$$

$$\frac{2}{n} \xrightarrow{n \rightarrow \infty} 0, \text{ since } a \text{ arbitrary}$$

$$= P(X=a) = 0 \quad \forall a \in (0,1) \quad \square //$$

(NO DISCRETE MODEL POSSIBLE)

Cumulative Distribution Function

X rv, cdf defined by:

$$F_X(x) = P_X((-\infty, x])$$

$$= P(\{\omega \in \Omega \mid X(\omega) \leq x\})$$

We shorten that to:

$$= P(X \leq x)$$

Example 1.5.3

Roll dice, $X = \text{roll}$

→ Space of $X = \{1, \dots, 6\}$

→ pmf is $P_X(i) = \frac{1}{6}, i = 1, \dots, 6$

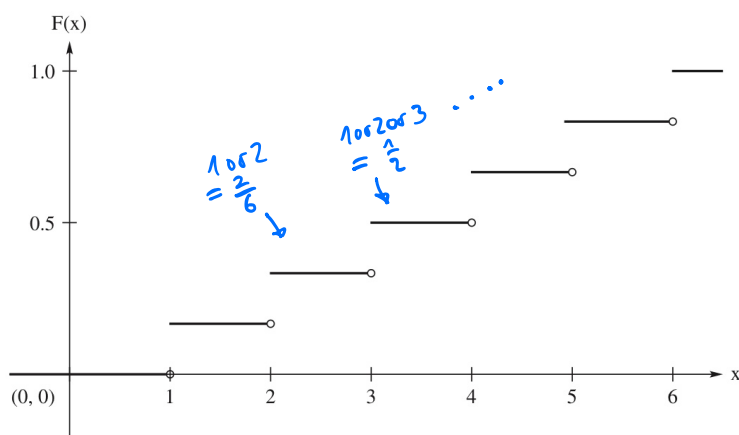
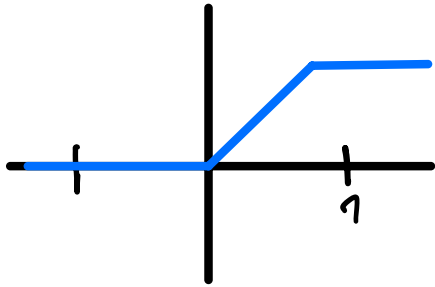


Figure 1.5.1: Distribution function for Example 1.5.3.

Example 1.5.4

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$



$$\begin{aligned} \leadsto F_Y(y) &= P(Y \leq y) \\ &= P(1-X \leq y) \\ &= P(X \geq 1-y) \\ &= 1 - (1-y) = y \end{aligned}$$

$$\rightarrow Y \stackrel{D}{=} X \text{ but } Y \neq X$$

Theorem 1.5.1

$$a) \forall a, b : a < b \Rightarrow F(a) \leq F(b)$$

$$b) \lim_{x \rightarrow -\infty} F(x) = 0$$

$$c) \lim_{x \rightarrow \infty} F(x) = 1$$

$$d) \lim_{x \rightarrow x_0} F(x) = F(x_0)$$

Proof a)

$$a < b \Rightarrow \{X \leq a\} \subset \{X \leq b\}$$

→ Result follows from

monotony.

d) Let $\{x_n\}$ be sequence $\in \mathbb{R}$

s.t. $x_n \rightarrow x_0$. Let $C_n = \{X \leq x_n\}$

$$\text{Then } \bigcap_{n=1}^{\infty} C_n = \{X \leq x_0\} := \{C_n\}$$

Sequence is decreasing,

hence:

$$\lim_{n \rightarrow \infty} F(x_n) = P\left(\bigcap_{n=1}^{\infty} C_n\right) = F(x_0)$$

□ //

Theorem 1.5.2

X be r.v. Cdf be F_X .

For $a < b$:

$$P(a < X \leq b) = F_X(b) - F_X(a).$$

Proof

$$\{-\infty < X \leq b\} = \{-\infty < X \leq a\} \cup \{a < X \leq b\}$$

Example 1.5.5

X be lifetime, X cdf:

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-x}, & 0 \leq x \end{cases}$$

pdf of $X \rightarrow \frac{d}{dx} F_X(x)$ (unsteady in 0)

$$f_X(x) = \begin{cases} e^{-x}, & 0 < x < \infty \\ 0, & \text{elsewhere} \end{cases}$$

$$\begin{aligned} P(1 < X \leq 3) &= F_X(3) - F_X(1) \\ &= \int_1^3 e^{-x} dx \end{aligned}$$

$$= e^{-1} - e^{-3} = 0,318 //$$

Theorem 1.5.3 $\forall$ r.v X

$$P(X=x) = F_X(x) - F_X(x-)$$

$$\forall x \in \mathbb{R} : F_X(x-) = \lim_{z \rightarrow x} F_X(z)$$

Proof: for any $x \in \mathbb{R}$:

$$\{x\} = \bigcap_{n=1}^{\infty} \left(x - \frac{1}{n}, x\right]$$

$\Rightarrow \{x\}$ is limit of decreasing sequence of sets. Hence:

$$P(X=x) = P\left(\bigcap_{n=1}^{\infty} \left\{x - \frac{1}{n} < X \leq x\right\}\right)$$

$$= \lim_{n \rightarrow \infty} P\left(x - \frac{1}{n} < X \leq x\right)$$

$$= \lim_{n \rightarrow \infty} (F_X(x) - F_X(x - \frac{1}{n}))$$

$$= F_X(x) - F_X(x-) \quad \square //$$

Example 1.5.6

X have cdf:

$$F_X(x) = \begin{cases} 0, & x < 0 \\ \frac{x}{2}, & 0 \leq x < 1 \\ 1, & 1 \leq x \end{cases}$$

Then:

$$P(-1 < X \leq \frac{1}{2}) = F_X(\frac{1}{2}) - F_X(-1)$$

$$= \frac{1}{4} - 0 = \frac{1}{4}$$

AND

$$P(X=1) = F_X(1) - F_X(1-)$$

$$= 1 - \frac{1}{2}$$

$$= \frac{1}{2} //$$

Example 1.5.7

Suppose X pmf

$$P_X(x) = \begin{cases} cx, & x=1, \dots, 10 \\ 0, & \text{elsewhere} \end{cases}$$

Then,

$$1 = \sum_{x=1}^{10} P_X(x)$$

$$= \sum_{x=1}^{10} cx$$

$$= c(1 + \dots + 10)$$

$$= 55c$$

$$\Rightarrow c = \frac{1}{55} //$$

Example 1.5.8

Suppose :

$$f_X(x) = \begin{cases} cx^3, & 0 < x < 2 \\ 0, & \text{elsewhere} \end{cases}, \quad c \in \mathbb{R}$$

$$\rightarrow \int_0^2 cx^3 dx$$

$$= c \cdot \left[\frac{1}{4} x^4 \right]_0^2$$

$$= \underline{4c}$$

$$\Rightarrow c = \frac{1}{4}$$

So for:

$$P\left(\frac{1}{4} < X < 1\right) = \int_{\frac{1}{4}}^1 \frac{1}{4} x^3 dx$$

$$= \frac{255}{4096}$$

$$= 0,06226 //$$